Supplier shortlisting with evidence you can check

Track 1 — Supplier Shortlisting, plus a complete Track 3 — Supply-Risk Scenario Planning.

Reading 23 supplier profiles by hand is slow. Trusting a tool that just names a winner is worse.

A buyer has **23 supplier profiles** and **7 mandatory requirements**. Working out who qualifies means reconciling certifications, order minimums, quality history and lead times across every document.

But a tool that answers "Supplier 1" and nothing else is worse than no tool. It is confident and unverifiable — and someone places a real order on it.

The model reads.
The code decides.

The LLM never compares a number to a threshold. It reports what a document says, and quotes it. Code does the arithmetic — and shows it.

```
5000 units ≤ 5000 units
```

- A supplier sitting **exactly on a limit** is settled by an operator, not by a model's judgement on a borderline case.
- **Every verdict shows its arithmetic** — checkable in a second, by someone who does not trust you.
- **Exploring a constraint is free.** Drag the buyer's order-quantity ceiling and all **161** verdicts re-decide in the browser — no model call, nothing re-read.
- Every citation is **verified byte-for-byte** against the source document. A model cannot talk its way past a string comparison.

> One finding that only exists because recompute is instant: raising the order-quantity limit alone changes nothing, and raising the lead-time limit alone changes nothing — but move both and a supplier qualifies. No single constraint was the obstacle.

MEASURED AGAINST 161 HAND-AUTHORED LABELS	THIS SYSTEM	RULE-BASED BASELINE
Accuracy	98.1%	91.9%
Pre-registered labels only (94)	96.8%	88.3%
Critical errors	0	12
Citation correctness	100%	–
Hallucinated quotes	0	–

> The accuracy gap is modest. The **critical-error** gap is the one that matters — six of the baseline's twelve were false passes, the mistake a buyer acts on.

THE OPPONENT IS REAL

The baseline is not a strawman

NegEx negation detection, explicit-absence detection, certificate date arithmetic. My first version was a strawman; research showed a competent 200-line script would match it, so I rebuilt it.

THE EXAM CAME FIRST

94 labels pre-registered

Ninety-four of 161 expected verdicts were written down in the data manifest before those documents ever met a model. That subset is scored separately — it cannot be flattered by hindsight.

THE CHECKER WAS CHECKED

92 corrupted citations, 92 caught

A zero hallucination rate means nothing if the detector cannot detect. Against 92 deliberately corrupted citations it caught all 92, and rejected none of the genuine ones.

People under-scrutinise confident automated recommendations. The failure mode worth designing against is not the system being wrong — it is the system being wrong and **sounding right**.

- **A counter-argument panel that is always on** — who is cheaper, where the winner is weak, how small a change flips it.
- **No action controls anywhere** — no approve, no contact, no place order. The system produces a shortlist and a stopping point.
- **Abstention is a real answer.** Missing evidence returns `insufficient-evidence`, never a guess.
- **Model confidence is deliberately not displayed.** Of 16 verdicts where the system declined to decide, 11 reported high confidence.

- What each requirement **costs you** — with the caveat beside the saving, not in a footnote.
- Who takes over if the **top supplier cannot deliver**.
- What a **25% lead-time slip** does: four eligible suppliers become two.
- Whether the order can be **split** — order minimums enforced, concentration measured as HHI.

THE FINDING

The constraint was never the order size

Dual sourcing at launch volume looked impossible: at the ratios procurement teams actually use, the second supplier's share fell below every candidate's minimum order quantity.

Then one small-batch manufacturer entered the set and it became possible at every ratio. The binding constraint was the **supplier set**. The tests were rewritten to assert the arithmetic, not the conclusion.

Three errors in 161. All of them the same bug.

It reports `CONFLICTING` when a document gives different values for different *scopes* — a certificate held by a sister site, order minimums listed per product line, an Indian head office with a Sri Lankan factory.

Those are not contradictions. It should select the value that applies. It correctly flagged the one genuine self-contradiction, and correctly left the consistent-restatement control alone.

> All three block the supplier and route to a human, so they fail in the safe direction. They are still errors, and fixing the extraction schema to carry applicability is the first thing I would build.

- **Fix the conflict over-reporting** — the extraction schema needs a notion of applicability, so the model can select rather than surrender.
- **Get a second annotator.** One person wrote the corpus, the labels and the system. Inter-annotator agreement would establish the ceiling; right now nobody knows it.
- **Test on real supplier PDFs.** This corpus is uniformly formatted because I generated it, which flatters pattern matching and understates the baseline gap.
- **Landed cost.** Freight, duties and Incoterms appear twice in the entire corpus. Every saving reported here is a unit-price saving, and it says so.

254	0	161
ASSERTIONS, TEN SUITES	API KEYS TO RUN IT	VERDICTS, EVERY ONE CITED